

Methods to optimize tribological properties of pineapple leaf fiber epoxy composites

Borhen Louhichi^a, Djuansjah Joy^g, Santosh Kumar Sahu^{b,*}, Nadir Ayrilmis^c,
It Ee Lee^{d,e,**}, Eng Eng Ngu^{d,f}, Harishbabu Sundarasetty^b

^a Engineering Sciences Research Center (ESRC), Deanship of Scientific Research, Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh 11432, Saudi Arabia

^b School of Mechanical Engineering, VIT-AP University, Besides A.P. Secretariat, Amaravati, Andhra Pradesh 522237, India

^c Department of Wood Mechanics and Technology, Faculty of Forestry, Istanbul University-Cerrahpasa, Bahcekoy, Sariyer, Istanbul 34473, Turkey

^d Faculty of Artificial Intelligence and Engineering, Multimedia University, Cyberjaya 63100, Malaysia

^e Centre for Smart Systems and Automation, COE for Robotics and Sensing Technologies, Multimedia University, Cyberjaya 63100, Malaysia

^f Centre for Electric Energy and High Voltage Engineering (CEEHVE), COE for Robotics and Sensing Technologies, Multimedia University, Cyberjaya 63100, Malaysia

^g College of Engineering, Imam Mohammad Ibn Saud Islamic University (IMSIU), Riyadh 11432, Saudi Arabia

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ABSTRACT

This study explores the enhancement of tribological properties in epoxy composites reinforced with pineapple leaf fibre (PALF) (*Ananas comosus*) using optimization and machine learning techniques. The rationale behind this research is to develop sustainable, high-performance materials for industrial applications by utilizing bio-based fibers, which offer environmental benefits. Response Surface Methodology (RSM) with Central Composite Design (CCD) was used, involving 24 runs. Key factors included fiber weight percentage (0 wt% pure epoxy and 30 wt% PALF/epoxy), applied load (10 N and 30 N), sliding velocity (0.7 mm/s and 2 mm/s), and sliding distance (500 mm and 1500 mm). X-ray diffraction (XRD) analysis showed that pure epoxy exhibited an amorphous structure, while the 30 wt% PALF/epoxy composite displayed a peak shift to 23.9°, indicating increased crystallinity. Analysis of Variance (ANOVA) revealed that increasing fiber content improved tribological properties. The composite with 30 wt% PALF at optimized condition showed a 94 % reduction in wear rate and a 78 % decrease in the coefficient of friction (COF), with R^2 values of 0.9135 for wear rate and 0.9203 for COF. Three machine learning models—linear regression, gradient boosting, and Gaussian Process (GP) regression—were employed for predictions. The GP model outperformed the others, achieving R^2 values above 0.98. SHAP analysis identified fiber weight percentage as the most influential factor on wear rate. This study demonstrates that experimental, statistical, and machine learning methods optimize tribological properties of bio-composites for industrial applications.

1. Introduction

The growing awareness of environmental concerns has led to increased adoption of sustainable materials across industries such as construction, packaging, and the automotive sector (Skosana et al., 2025; Chee et al., 2019). Traditional thermoset polymers, often derived from crude oil, are nonrenewable resources that pose significant environmental challenges. This has driven research into developing eco-friendly composite materials by reinforcing thermoset with natural fibers, thus supporting the principle of reducing reliance on non-renewable resources while maintaining high performance (Feng

et al., 2026). Epoxy is a type of thermoset polymer known for its excellent mechanical properties, high strength, chemical resistance, and durability. It remains a popular matrix material for composites (Maleki et al., 2025). These properties make epoxy composites ideal for applications requiring robust performance under stress, including friction and abrasion. Wear testing is widely used to assess the durability of these composites, specifically evaluating their ability to withstand friction, abrasion, and mechanical stress. Such testing is essential to confirm that bio-based composites can substitute a portion of petroleum-based materials while still meeting environmental sustainability goals (Tasgin et al., 2024).

* Corresponding author.

** Corresponding author at: Faculty of Artificial Intelligence and Engineering, Multimedia University, Cyberjaya 63100, Malaysia

E-mail addresses: santoshkumar.sahu@vitap.ac.in (S.K. Sahu), ielee@mmu.edu.my (I.E. Lee).

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Numerous studies have explored the use of various fibers to enhance the wear resistance of epoxy composites. Chavhan and Wankhade (2020) optimized steel-embedded E-glass fiber/epoxy composites, achieving a 50.47 % reduction in specific wear rate (SWR) and a 31.48 % decrease in the coefficient of friction (COF) with 10 % filler content. Venkatesh et al. (2023) found that the 70E/10KF/20SF composite exhibited higher tensile and flexural strength, with a low wear rate of 0.012 mg/min. Sumesh et al. (2024) examined various natural fiber-reinforced epoxy composites (NFRCS), including cotton (*Gossypium hirsutum*), sisal (*Agave sisalana*), coir (*Cocos nucifera*), and wool, showing that cotton (30 wt%) had the highest tensile strength (52.81 MPa) and thermal conductivity (1.017 W/mK). Singh et al. (2021) reinforced silica nanoparticles into sisal fiber in epoxy, improving tensile strength and impact resistance. Giridharan et al. (2025) reinforced epoxy composites with pineapple leaf fiber (*Ananas comosus*) and roselle (*Hibiscus sabdariffa*), finding 30 % PALF and 20 % roselle exhibited the highest wear resistance and mechanical strength. Mahmud et al. (2025) showed that natural fibers like jute (*Corchorus olitorius*), hemp (*Cannabis sativa*), and sisal (*Agave sisalana*) improved tensile strength by 50 % and heat stability by 20 %. Other studies, such as those by Muralidhara et al. (2025), Dharmalingam and Sasikumar (2025), reported significant improvements in wear resistance and thermal stability with the addition of halloysite nanotubes (HNTs) and nanosilica. Furthermore, Lakshmi pathi et al. (2025) developed composites using *Amaranthus dubius* fiber and biosilica particles, exhibiting excellent wear resistance, while Manoj et al. (2025) developed composites with waste sorghum husk (*Sorghum bicolor*) ash and biochar, demonstrating enhanced composite performance with low thermal conductivity and COF. Saran and Satapathy (2025) explored glass dust as a filler in hemp fiber (*Cannabis sativa*) composites, finding the lowest wear rate at 20 wt% filler, a sliding distance of 250 m, and a 10 N load.

Previous studies have focused on enhancing the mechanical and tribological properties of epoxy composites with natural fibers and biofillers. However, optimization using advanced statistical techniques such as Response Surface Methodology (RSM), ANOVA, and machine learning has been limited. This research introduces a novel hybrid approach that combines optimization techniques with machine learning models to both optimize and predict the performance of epoxy composites reinforced with pineapple leaf fiber (PALF) (*Ananas comosus*). PALF was selected for its exceptional mechanical properties, biodegradability, and availability as an agricultural byproduct, making it an ideal candidate for eco-friendly composites. The novelty of this study lies in integrating advanced optimisation techniques with machine learning models—such as Linear Regression (LR), Gradient Boosting, and Gaussian Process (GP)—to predict tribological behaviour and optimise key parameters, including fibre weight percentage, applied load, sliding velocity, and sliding distance. By leveraging these machine learning techniques, the study improves the accuracy of predicting wear resistance and coefficient of friction. The objectives of the study include optimising the performance of PALF-epoxy composites, evaluating the impact of various testing parameters on wear resistance and coefficient of friction, assessing the performance of PALF composites, and demonstrating PALF's potential as a sustainable reinforcement for high-performance composites.

2. Materials and methods

2.1. Materials

Epoxy resin (LY 556) and hardener (HY 951) were sourced from Covai Seenu & Co. Ltd. (Coimbatore, India). The resin had a viscosity range of 10,000–12,000 mPa·s and a density between 1.15 g/cm³ and 1.20 g/cm³, while the hardener's density ranged from 0.97 g/cm³ to 1.00 g/cm³. By-directional pineapple leaf fibers (PALF) (*Ananas comosus*) in sheet form, with an average fiber length of 30–50 mm and a moisture content of less than 10 %, were obtained from a local vendor in

Vijayawada, India as reinforcement material for composite fabrication.

2.2. XRD

X-ray diffraction (XRD) analysis is essential for determining the crystal structure and phase composition of materials. The measurements were carried out using a Rigaku Miniflex600-C instrument, employing Cu K α radiation (wavelength = 1.54 Å). The scan was performed over a 2 θ range from 5° to 80°.

2.3. Experimental design

Response Surface Methodology (RSM) was employed to optimize the process response variables, given its ability to simultaneously explore a large number of factors with minimal bias (Singh et al., 2025). A Central Composite Design (CCD) with 24 random runs was selected for this study, as it is particularly effective in fitting quadratic models and capturing both linear and second-order effects of the factors. The variables considered in the optimization process included applied pressure (N), fiber weight percentage (wt%), sliding speed (mm/s), and sliding distance (mm) (Kumar et al., 2024; Sundarasetty and Sahu, 2025). The values for these parameters are provided in Table 1. The use of randomization in the runs further enhanced the accuracy and reliability of the optimization, helping minimize potential bias in the results (Jagadeesan et al., 2022).

2.4. Composite Preparation

Fig. 1 illustrates the overall process for fabricating the epoxy/pineapple leaf fiber (PALF) (*Ananas comosus*) composite using the hand layup method. Initially, the PALF sheet was dried in an oven at 60 °C for 4 h to reduce its moisture content and weighed to achieve the required weight percentage in the composite. Following this, the total composite weight was calculated, and the epoxy resin and hardener were weighed accordingly to maintain the correct 10:1 wt ratio between the resin and hardener, as per the vendor's instructions. The epoxy resin and hardener were then thoroughly mixed for 3–5 min to ensure complete incorporation of the components. A release agent was sprayed onto the molds to prevent the composite from sticking during curing. The bidirectional PALF fibers were carefully placed into the mold using the hand layup technique, and the resin mixture was applied using a squeegee to ensure full saturation of the fibers. The composite was left to cure at room temperature for 24 h. Afterward, it was post-cured in an oven at temperatures between 60°C and 80°C for 2–4 h to ensure complete curing of the resin. Once fully cured, the composite was cut into wear-testing samples in accordance with ASTM standards (Feng et al., 2024; Nagaraja et al., 2024).

2.5. Wear Testing

The pin-on-disc model DOCOM TR-20LE-PHM400-CHM400 test setup used to assess the tribological properties of the epoxy and pineapple leaf fiber (PALF)/ epoxy composites in accordance with ASTM G99 standards (Salguero et al., 2022). The setup includes a steel disc (EN 31) counterface that serves as the opposing surface during wear testing. Key parameters, including applied load, sliding velocity, sliding distance, and fiber weight percentage (wt%), were evaluated through

Table 1
Process Factors and Levels for Response Surface Methodology.

S.No	Factors	Units	Minimum	Maximum
1	Fiber (B ₁)	wt%	0 (Pure Epoxy)	30
2	Applied Load (B ₂)	N	10	30
3	Sliding Velocity (B ₃)	mm/s	0.7	2
4	Sliding Distance (B ₄)	mm	500	1500

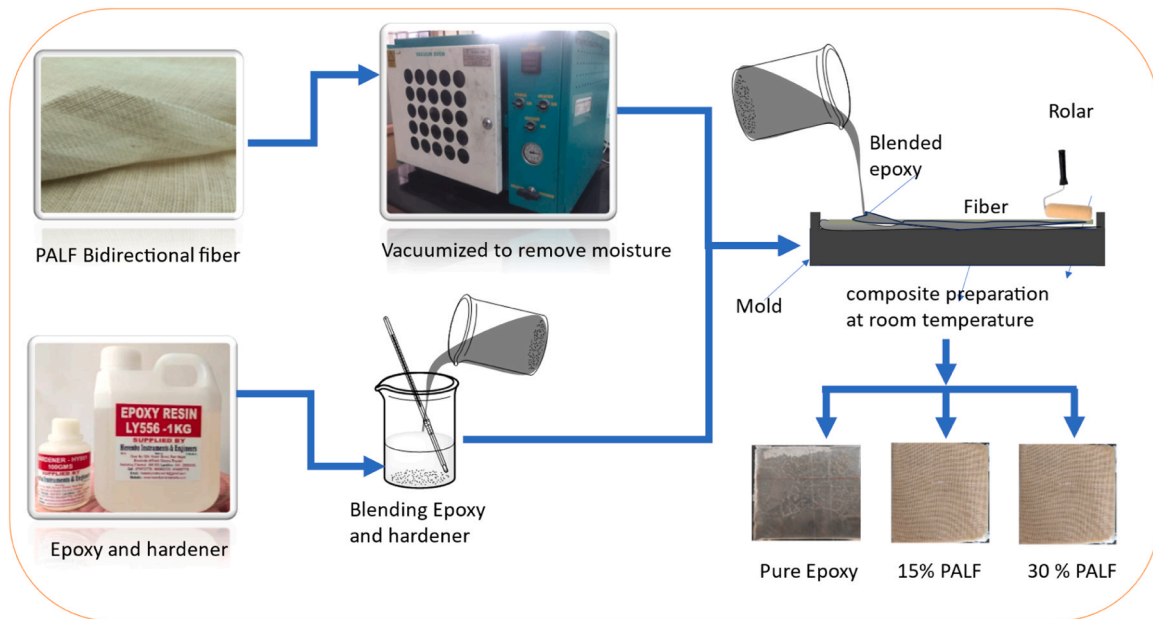


Fig. 1. Composite preparation process.

experiments designed using a face-centered second-order approach based on Response Surface Methodology (RSM), as outlined in Table 2. The volume of wear was determined from the experimental data by measuring the wear depth. Wear rate evaluation is performed using Eqs. (1–2) (Sidh and Hirani, 2025). Additionally, by applying Eq. (3) (Harishbabu et al., 2025a) along with the frictional force data obtained from the tests, the tribological performance of the composite can be assessed by calculating the friction coefficient under the specified testing conditions.

$$\text{Wear rate} = \frac{\text{Volume of the material lost (mm}^3\text{)}}{\text{Applied load (N)} \times \text{Sliding distance (mm)}} \quad (1)$$

$$\text{Volume of worn} = \text{Area of wear (mm}^2\text{)} \times \text{height (mm)} \quad (2)$$

Table 2

Experiments designed by response surface methods.

Exp. No.	Fiber (B ₁) (wt%)	Applied Load (B ₂) (N)	Sliding Velocity (B ₃) (mm/s)	Sliding Distance (B ₄) (mm)
S1	30	10	0.7	1500
S2	0	10	2	1500
S3	0	10	0.7	1500
S4	15	10	1.35	1000
S5	0	10	0.7	500
S6	0	20	1.35	1000
S7	30	30	2	1500
S8	15	20	1.35	500
S9	15	30	1.35	1000
S10	30	10	2	1500
S11	30	30	2	500
S12	0	30	2	500
S13	0	10	2	500
S14	30	20	1.35	1000
S15	15	20	2	1000
S16	15	20	0.7	1000
S17	30	30	0.7	1500
S18	15	20	1.35	1500
S19	30	10	0.7	500
S20	30	30	0.7	500
S21	0	30	0.7	1500
S22	0	30	0.7	500
S23	0	30	2	1500
S24	30	10	2	500

$$\text{Coefficient of friction} = \frac{\text{Friction force (N)}}{\text{Applied load (N)}} \quad (3)$$

2.6. Optical microscopy

The Zeiss Primotech optical microscope was used to examine the wear patterns on the samples. This microscope enables us to observe wear tracks and scratches on the surface, providing a detailed view of surface degradation.

2.7. Statistical analysis

2.7.1. Design of experiments

The experiments were designed using a Central Composite Design (CCD) within the framework of Response Surface Methodology (RSM) to optimize the tribological properties of epoxy composites as outlined in Table 2. This approach was chosen for its ability to assess the effects of multiple factors and their interactions while minimizing the number of experimental trials to just 24 runs, using a quadratic model. The strategy incorporated a randomized research design to ensure that the results accounted for variability across all experimental conditions. This design enabled the identification of both linear and nonlinear effects of parameters such as fiber weight percentage, applied load, sliding velocity, and sliding distance on tribological properties, offering valuable insights into the process-property relationship and providing a clear framework for optimizing composite performance (Harishbabu et al., 2025a).

2.8. Machine learning

The tribological behaviour of pure epoxy and pineapple leaf fibre (PALF) (*Ananas comosus*) reinforced epoxy composites was predicted using machine learning (ML) techniques, including linear regression, gradient boosting, and Gaussian process regression. These models were trained to assess the impact of experimental factors, such as applied load, sliding velocity, sliding distance, and fiber weight percentage, on key wear-related metrics, such as friction coefficient and wear rate. To optimize the prediction performance, Hypergrid Search CV (Hyperparameter Grid Search) was used to fine-tune the models. The ML algorithms were evaluated for their ability to capture both linear and nonlinear interactions between the input factors and the material's

tribological properties. The performance of the models was assessed using several statistical metrics, including coefficients of determination (R^2), mean squared error (MSE), root mean squared error (RMSE), relative RMSE, mean absolute error (MAE), and mean absolute percentage error (MAPE), as outlined in Eqs. (4) to (9) (Ul-Hamid, 2021; Sundarasetty et al., 2025; Pernica et al., 2022).

$$R^2 = 1 - \frac{\sum_{i=1}^k (e_i - \bar{e}_i)^2}{\sum_{i=1}^k (e_i - \bar{e})^2} \quad (4)$$

$$MSE = \frac{1}{k} \sum_{i=1}^k (e_i - \bar{e}_i)^2 \quad (5)$$

$$RMSE = \sqrt{MSE} = \sqrt{(e_i - \bar{e}_i)^2} \quad (6)$$

$$RRMSE = \frac{RMSE}{\bar{e}} = \frac{\sqrt{(e_i - \bar{e}_i)^2}}{\bar{e}} \times 100 \quad (7)$$

$$MAE = \frac{1}{k} \sum_{i=1}^k |e_i - \bar{e}_i| \quad (8)$$

$$MAPE = \frac{1}{k} \sum_{i=1}^k \left| \frac{e_i - \bar{e}_i}{e_i} \right| \times 100 \quad (9)$$

Here, 'k' denotes the overall count of observations or trials. ' e_i ' stands for the numerical value actually measured during the k^{th} trial, while ' $\bar{e}$ ' represents the average of all numerical values from all trials.

2.8.1. Linear regression (LR)

A linear regression model was employed to analyze the linear relationship between independent variables, such as fiber weight percentage, applied load, sliding velocity, and sliding distance, and the dependent variables (target variables), such as wear rate and coefficient of friction. In this study, linear regression was applied to optimize the tribological properties of the epoxy-PALF composite. As a basic yet effective model, linear regression provides initial predictions and serves as a benchmark for comparison, assuming a linear correlation between the input factors and output responses. To ensure accurate results, the dataset was divided into training and testing subsets. The mathematical formulation representing the relationship between the dependent and independent variables is provided in Eqs. (10) and (11) (Silhavy et al., 2017).

Linear Regression model,

$$\bar{T} = \varnothing_0 + \varnothing_1 B_1 + \varnothing_2 B_2 + \varnothing_3 B_3 + \varnothing_4 B_4 \quad (10)$$

Here, B_1 , B_2 , B_3 , and B_4 are the processing and testing parameters, respectively, and $\varnothing_0$, $\varnothing_1$, $\varnothing_2$, $\varnothing_3$, and $\varnothing_4$ are the model coefficients.

Coefficients of the model,

$$\bar{\varnothing} = (B^T B)^{-1} B^T T \quad (11)$$

Here, B is the scaled matrix of processing parameters, and T is the scaled responses.

2.8.2. Gradient boosting

A gradient boosting model is used to explore the intricate relationships between independent variables such as fiber weight percentage, applied load, and sliding conditions, along with dependent variables like wear rate and coefficient of friction. This model is highly effective in optimizing the tribological performance of the epoxy and PALF/epoxy composite, as it captures the non-linear relationships between these variables. Gradient boosting works by iteratively adding weak learners, typically decision trees, to the model. Each new learner corrects the

mistakes of the previous one, gradually improving prediction accuracy. This process of staged learning is crucial for the model to identify and learn complex patterns in the data. To enhance the model's reliability, the dataset is split into training and testing subsets, with predictions becoming more refined as weak learners are adjusted, as illustrated in Eqs. (12)–(15) (Qiuqian et al., 2025).

Define a dataset by

$$D = \{((B_i, T_i))\}_{i=1}^m \quad (12)$$

Here, B_i is [fiber wt%, applied load, sliding velocity, sliding distance], and here, T_i is [wear rate, coefficient of friction].

$$F_M(B) = \sum_{m=1}^M L_m h_m(B) \quad (13)$$

For the Gradient descent update,

$$F_m(B) = F_{m-1}(B) + R_m h_m(B) \quad (14)$$

Here L_m is the learning rate or weight applied to each tree, M is the total number of iterations (trees), and $h_m(B)$ is the m^{th} weak learner (decision tree).

h_m is trained to fit the pseudo-residuals

$$h_m \approx r_i^m = \left[\frac{\partial l(G_i, F_{m-1}(B_i))}{\partial F_{m-1}(B_i)} \right] \quad (15)$$

Here r_i^m is the pseudo-residual for data point i at boosting iteration m, G_i is the actual target value, $F_{m-1}(B_i)$ is the Prediction from the model after m-1 iterations, and $\partial l(G_i, F_{m-1}(B_i))$ is the loss function (e.g., mean squared error).

Here, l is considered a smooth loss function that quantifies the error between the predicted and actual values. Each f_a represents a decision tree added during iteration a . To avoid overfitting, Eq. 14 introduces a regularization term. $N(f)$, where T is the number of terminal nodes (leaves), ω is the score assigned to each leaf, and β and δ are constants that penalize excessive model complexity.

2.8.3. Gaussian process (GP)

A Gaussian Process (GP) regression model is used to optimize the tribological performance of epoxy-PALF composites by capturing the nonlinear relationships between variables. Unlike traditional models that assume a predefined functional form, GP regression defines a distribution over all possible functions, offering a more flexible and robust approach to modeling complex, uncertain processes. The GP model is governed by a kernel function (mean function) and a covariance function, which represent assumptions about the data's smoothness and variability. This flexibility allows the model to effectively adapt to data with inherent uncertainty, making it especially useful when the true functional relationship between variables is difficult to explicitly define. To improve the model's robustness, the dataset is split into training and testing subsets. The model's predictions are iteratively refined as it continuously updates its understanding of the data's structure based on observed outcomes. This iterative process enables GP to generate accurate predictions that account for both data uncertainty and complex dependencies, as detailed in Eqs. (16) and (17) (Harishbabu et al., 2025b; Zhang et al., 2025).

$$f(B) \sim GP(0, K(B, B')) \quad (16)$$

$$\bar{T}(B_*) = K(B_*, B) [K(B, B') + \alpha_n^2 I]^{-1} T \quad (17)$$

Here, $f(B)$ - Latent function representing the true mapping between inputs and outputs, $K(B, B')$ - Covariance matrix defined by the RBF kernel, B- Matrix of input variables (scaled processing and testing parameters), B_* - New input sample. $\bar{V}(B_*)$ -Predicted output (tribological property) for B_* , T: Observed output vector (scaled mechanical properties), α_n^2 - Noise variance, I- Identity matrix.

3. Results and discussion

3.1. X-ray diffraction (XRD)

The comparison of XRD patterns of pure epoxy and 30 wt% of PALF/epoxy composite samples are presented in Fig. 2. The x-axis represents the 2θ angle (in degrees). At the same time, the y-axis shows the intensity (in arbitrary units, a.u.) of the diffraction peaks. For pure epoxy (black curve), a sharp peak is observed around 19° , indicating its amorphous nature and the absence of a significant crystalline structure. In contrast, the XRD pattern of the epoxy composite with 30 wt% PALF (red curve) reveals a shift of the main peak to around 23.9° , suggesting that PALF incorporation altered the structural order of the composite. Furthermore, a new peak was observed in the composite sample within the $35\text{--}45^\circ$ 2θ range, which is broader than the sharp peak seen in pure epoxy. This broad peak suggests that pineapple leaf fibers disrupt the regular packing of epoxy molecules, leading to a more disordered structure. These changes in peak positions and the appearance of a broadened peak in the composite imply that the PALF influences the epoxy's crystallinity and microstructure.

3.2. Wear test

The wear testing was conducted using a pin-on-disk test as per the experiment's design by the response surface method (RSM), and the experimentally obtained wear rate and COF are presented in Table 3, where the wear rates and COF values for each combination of factors are summarized.

3.2.1. Wear rate

Fig. 3 shows the experimentally obtained wear rate for the 24 samples listed in Table 3. The plot shows how variation in factors at three levels influences wear, with the x-axis representing the experimental number (Exp. No.) and the y-axis showing the wear rate. For instance, test condition S1, with a 30 wt% fiber, 10 N load, a sliding velocity of 0.7 mm/s, and a sliding distance of 1500 mm, has a wear rate of $0.72 \mu\text{m}^3/\text{N-mm}$, indicating relatively low wear under these conditions. In contrast, increasing the applied load and sliding velocity, as seen in test condition S9 (15 wt% fiber, 35 N load, 1.35 mm/s velocity, 1000 mm distance), results in a higher wear rate of $1.26 \mu\text{m}^3/\text{N-mm}$. Similarly, test condition S18, with a 15 wt% fiber, a 20 N load, a 1.35 mm/s sliding velocity, and a sliding distance of 1707 mm, exhibits the highest wear rate of $3.65 \mu\text{m}^3/\text{N-mm}$, due to a combination of a moderate applied load, sliding velocity, and sliding distance. Test condition S7 (30 wt%

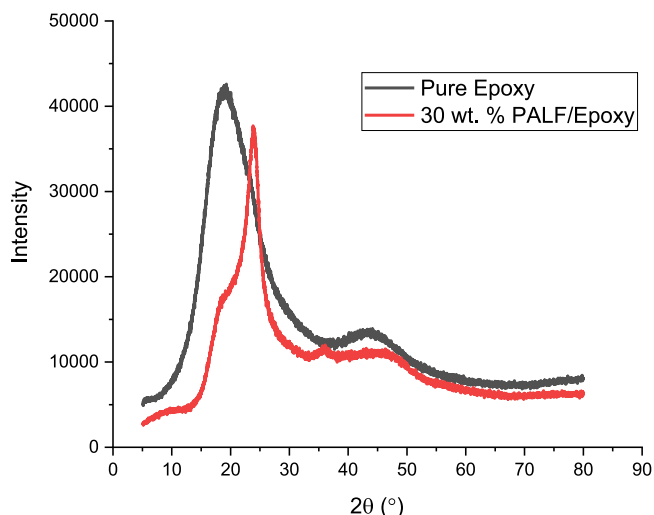


Fig. 2. XRD results of pure epoxy and 30 wt% PALF/epoxy composite sample.

Table 3

Summarised experimentally obtained Wear Rate and Coefficient of Friction (COF).

Exp. No.	Fiber (B ₁) (wt%)	Applied Load (B ₂) (N)	Sliding Velocity (B ₃) (mm/s)	Sliding Distance (B ₄) (mm)	wear rate $\mu\text{m}^3/\text{N-mm}$	COF
S1	30	10	0.7	1500	0.724	0.737
S2	0	10	2	1500	0.29	0.418
S3	0	10	0.7	1500	0.306	0.417
S4	15	10	1.35	1000	1.26	0.705
S5	0	10	0.7	500	0.171	0.08
S6	0	20	1.35	1000	1.3	0.361
S7	30	30	2	1500	0.263	0.462
S8	15	20	1.35	500	0.569	0.305
S9	15	30	1.35	1000	0.695	0.428
S10	30	10	2	1500	0.965	0.72
S11	30	30	2	500	0.93	0.563
S12	0	30	2	500	0.243	0.396
S13	0	10	2	500	0.18	0.341
S14	30	20	1.35	1000	0.233	0.165
S15	15	20	2	1000	0.762	0.454
S16	15	20	0.7	1000	0.506	0.679
S17	30	30	0.7	1500	0.235	0.683
S18	15	20	1.35	1500	0.812	0.78
S19	30	10	0.7	500	3.65	0.585
S20	30	30	0.7	500	1.24	0.415
S21	0	30	0.7	1500	0.448	0.487
S22	0	30	0.7	500	0.649	0.42
S23	0	30	2	1500	0.252	0.545
S24	30	10	2	500	2.15	0.565

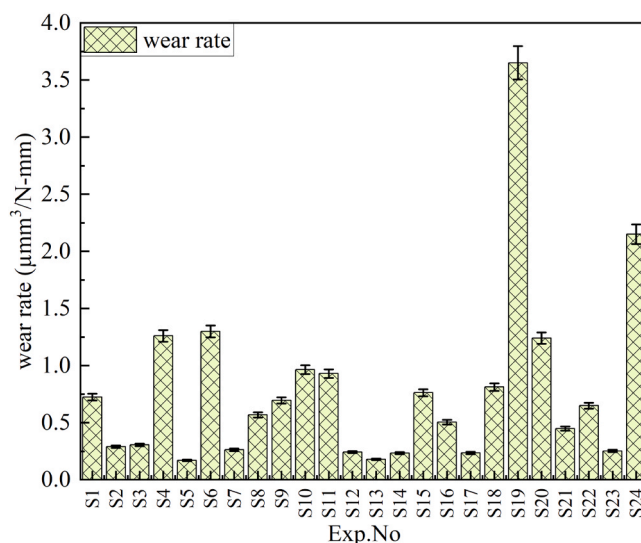


Fig. 3. Wear Rate vs. Experimental Conditions for Various Fiber wt%, Applied Load, Sliding Velocity, and Sliding Distance.

fiber, 30 N load, 2 mm/s velocity, 1500 mm distance) has a wear rate of $0.97 \mu\text{m}^3/\text{N-mm}$, where increased load and velocity also result in higher wear, but not as much as in test condition S18. On the other hand, S3 (0 wt% fiber, 10 N load, 0.7 mm/s velocity, 1500 mm distance) and test condition S5 (0 wt% fiber, 10 N load, 0.7 mm/s velocity, 500 mm distance) exhibit wear rates of $0.29 \mu\text{m}^3/\text{N-mm}$ and $0.31 \mu\text{m}^3/\text{N-mm}$, respectively, indicating lower wear under conditions of lower load and velocity. In summary, results show that maximum wear rate at 0 wt % fiber is $3.65 \mu\text{m}^3/\text{N-mm}$ (test condition S19) and minimum wear rate at 30 wt% fiber is 0.233 (test condition S14) demonstrated 94 % reduction in wear rate. The reason for this attributed to the increased reinforcement provided by the pineapple leaf fiber (PALF). The fiber improves the composite's structural integrity, distributing applied stresses more effectively, thereby reducing wear rate.

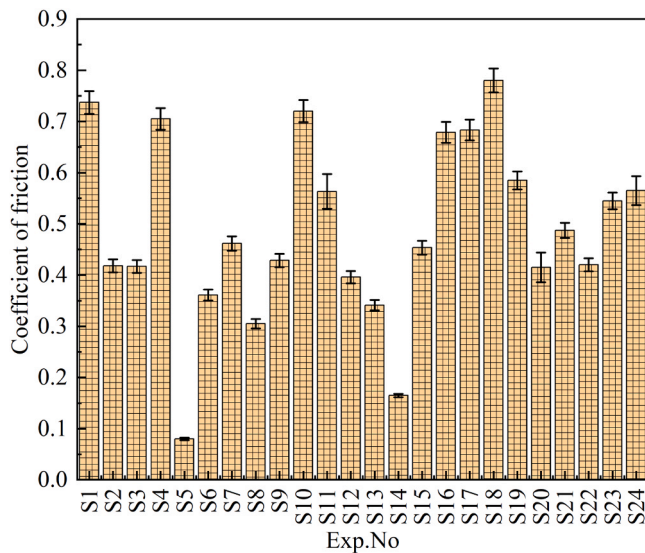


Fig. 4. Coefficient of friction vs. experimental conditions for different fiber wt %, applied load, sliding velocity, and sliding distance.

3.2.2. Coefficient of friction (COF)

Fig. 4 demonstrates that variations in testing parameters, including fiber wt%, applied load, sliding velocity, and sliding distance, affect the coefficient of friction. The plot with exp no. on the x-axis and COF on the y-axis. At test condition S1 has 30 wt% fiber content, with a 10 N load, a sliding velocity of 0.7 mm/s, and a sliding distance of 1500 mm, resulting in a coefficient of friction of 0.737, indicating relatively high friction under these conditions. In contrast, test condition S5 has 0 wt% fiber, with a 10 N load, a sliding velocity of 0.7 mm/s, and a sliding distance of 500 mm, yielding a coefficient of friction of 0.08, suggesting that lower fiber wt% and shorter sliding distance result in very low friction. Similarly, test condition S18, having 15 wt% fiber, 20 N load, 1.35 mm/s sliding velocity, and a sliding distance of 1707 mm, indicates a coefficient of friction of 0.78, reflecting the influence of a moderate applied load, sliding velocity, and sliding distance on increasing friction. On the other hand, test condition S6, having 0 wt% fiber, with a 20 N load, 1.35 mm/s sliding velocity, and a sliding distance of 1000 mm, gives the coefficient of friction as 0.361, while test condition S3 having 0 wt% fiber with a 10 N load, 0.7 mm/s sliding velocity, and 1500 mm sliding distance gives 0.418, suggest that friction increases with higher applied load and sliding velocity. The highest coefficient of friction is observed at test condition S17, with a coefficient of 0.78, which indicates increased fiber wt%, applied load, and sliding distance. The coefficient of friction tends to increase with applied load, sliding velocity, and sliding distance, though fiber wt% also significantly impacts friction behavior. In summary, results show that the maximum coefficient of friction (COF) at 0 wt% fiber is 0.737 (test condition S1) and the minimum COF at 30 wt% fiber is 0.165 (test condition S14), demonstrating a 78 % reduction in COF. The reason for this owing to lubrication effect of the pineapple leaf fiber (PALF). The fiber reinforcement reduces surface contact and promotes smoother sliding, leading to lower friction between the surfaces.

3.3. Optical microscopy

Fig. 5 illustrates optical microscopy images comparing the worn surfaces of pure epoxy and 30 wt% PALE/epoxy composite under

varying applied loads of 10 N, 20 N, and 30 N. In Fig. 5(a), the wear pattern for pure epoxy under a load of 10 N shows parallel shallow grooves and ploughing, a typical characteristic of mild wear. As the load increases to 20 N and 30 N, as shown in Fig. 5(b) and Fig. 5(c), the wear becomes more severe, with deeper grooves and ploughing, indicating adhesive wear (Ibrahim et al., 2024; Suresha et al., 2007). In contrast, the 30 wt% PALE/epoxy composite samples exhibit less surface damage under the same varying loads. At a 10 N load in Fig. 5(d), the surface shows some pits and a few scratches, but overall, the sample remains relatively undamaged. As the load increases to 20 N (Fig. 5(e)), some scratches appear, but large regions of the surface remain smooth with fewer grooves and pits. At the highest load of 30 N, shown in Fig. 5(f), abrasive wear becomes more noticeable, with visible scratches and fewer pits. The grooves in the composite are finer and less continuous than those observed in pure epoxy. This behavior aligns with previous studies suggesting that PALE fiber/matrix reinforcement enhances stiffness, facilitates load transfer, and mitigates severe grooving in PALE/epoxy composites (Kumar and Anand, 2019).

3.4. Statistical analysis by ANOVA

3.4.1. Wear rate

The ANOVA analysis was performed using response surface (RMS) based on a quadratic model with a Central Composite Design (CCD) to identify the most influential factor on wear rate in fiber wt%, applied load, sliding velocity, and sliding distance, as explained in Table S1. The overall model is significant, with an F-value of 8.12 and a p -value of 0.0011, indicating that it explains a substantial amount of variability in the data. Among the individual factors, Applied Load (B_2) has a significant effect on the outcome, with a p -value of 0.0091, while Fiber (B_1) has a marginal impact with a p -value of 0.05. However, sliding velocity (B_3) and sliding distance (B_4) were not significant, with p -values of 0.2369 and 0.5483, respectively. Numerous interactions are statistically significant, including the interaction between Fiber wt% and Applied Load ($B_1 \times B_2$) with a p -value of 0.0038, while the interaction between Fiber wt% and Sliding Distance ($B_1 \times B_4$) with a p -value of 0.0027, and the squared term interaction of Fiber wt% squared and Sliding Distance ($B_1^2 \times B_4$) with a p -value of 0.0442. These interactions signify that the combined effects of these factors influence the wear rate. In contrast, the interactions of sliding velocity and sliding distance ($B_3 \times B_4$) and the three-way interaction between fiber wt%, applied load, and sliding velocity ($B_1 \times B_2 \times B_3$) are found to be insignificant, with p -values of 0.1322 and 0.2802, respectively. The model explains 91.35 % of the variance ($R^2 = 0.9135$), demonstrating a strong fit and the adjusted R^2 of 0.8010, justifying the robustness of the model even after adjustment for the number of predictors. Eq. (18) represents a multiple regression model for predicting the wear rate based on the interaction effects and quadratic terms of the independent variables (B_1 , B_2 , B_3 , and B_4).

$$\begin{aligned} \text{Wearrate} = & (7.780 - 4.418B_1 - 2.489B_2 - 0.8449B_3 + 1.075B_4 - 3.268B_1 \\ & \times B_2 - 0.4928B_1 \times B_3 - 0.463B_1 \times B_4 + 1.197B_3 \\ & \times B_4 + 0.8335B_1 \times B_2 \times B_3 + 1.001B_1 \times B_3 \\ & \times B_4 - 0.7643 B_2 \times B_3 \times B_4 - 4.438 B_1^2 \times B_4 + 9.088 B_1 \\ & \times B_2^2) \times 10^{-7} \quad \dots \end{aligned} \quad (18)$$

3D surface plots in Fig. 6, illustrate the influence of parameters such as fiber wt%, applied load, sliding velocity, and sliding distance on wear rate. Fig. 6(a) presents the combined effect of fiber wt% and applied load (N) on wear rate. As fiber wt% varies from 0 % to 30 %, the wear rate

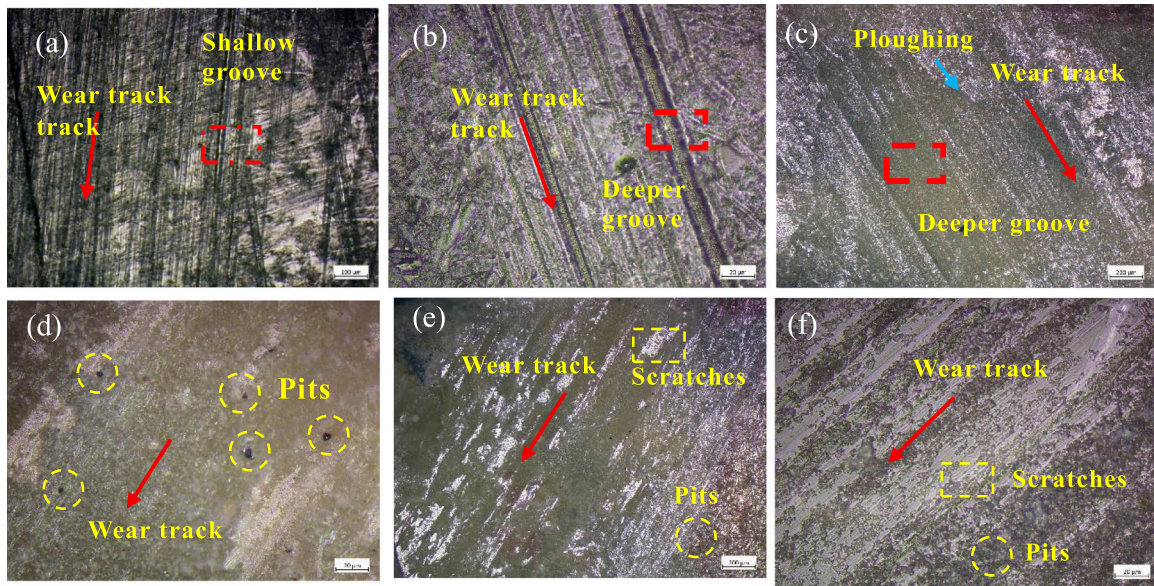


Fig. 5. Optical microscopy results at varying loads pure epoxy at a) 10 N, b) 20 N, c) 30 N loads; 30 wt% PALF/epoxy at d) 10 N, e) 20 N, f) 30 N loads.

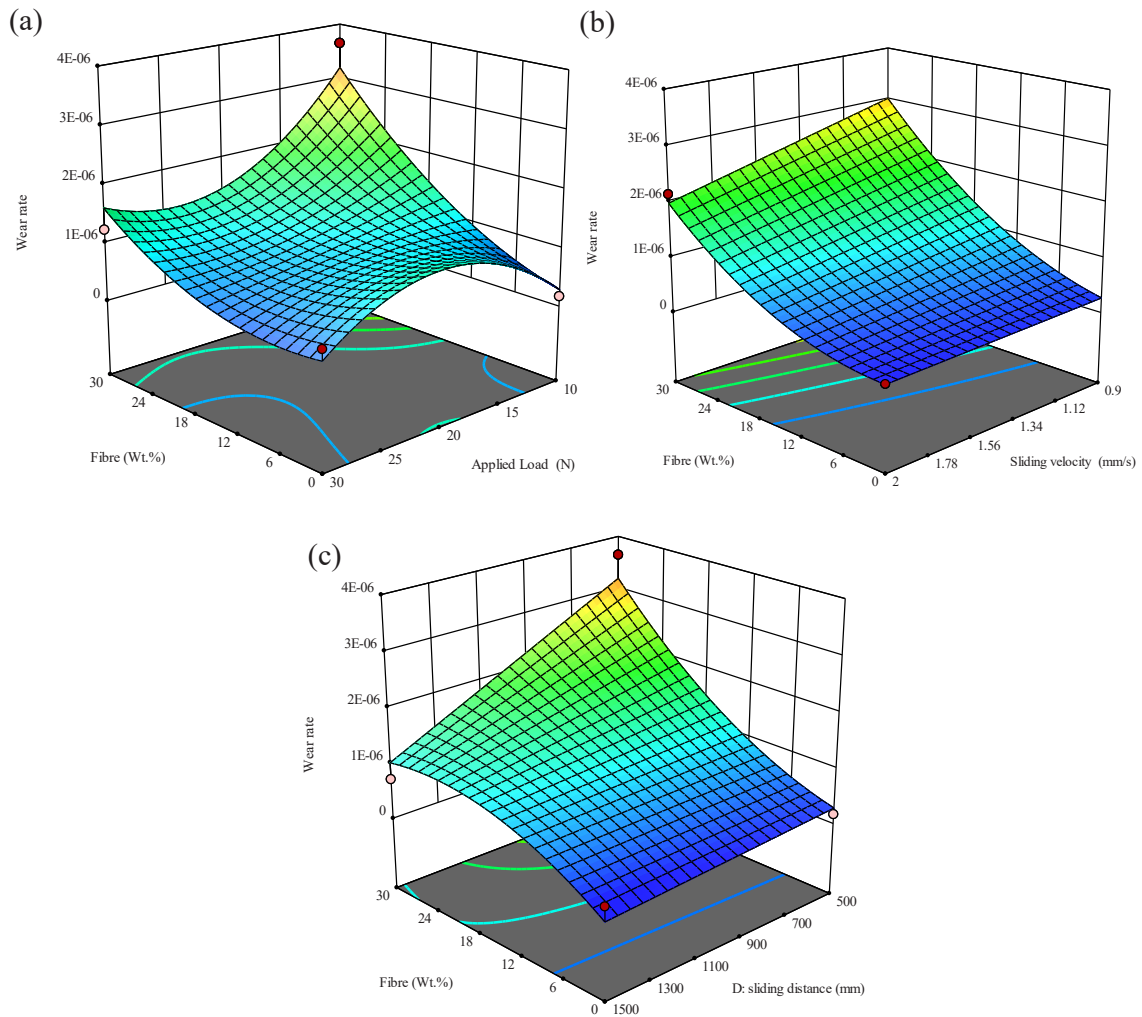


Fig. 6. 3D Surface Plots represent the effect on wear rate with (a) Fiber wt% vs Applied Load, (b) Fiber wt% vs Sliding Velocity, and (c) Fiber wt% vs Sliding Distance on Wear Rate.

enhances from 1.5×10^{-6} mm to 4.5×10^{-6} mm, especially when the applied load increases up to 20 N. This confirms a positive correlation between fiber wt% and applied load, both of which are directly proportional to the wear rate. Fig. 6(b) explores the integration of sliding velocity (m/s) with fiber wt%. As the fiber wt% rises from 0% to 30%, the wear rate increases from 1.2×10^{-6} mm to 4.0×10^{-6} mm. However, the increase in wear rate slows and decelerates when the sliding velocity exceeds 1 m/s. This proves that with higher fiber wt%, the contribution of sliding velocity is marginal to wear rate. Fig. 6(c) shows the interaction of fiber wt% with sliding distance (mm). For a 30% fiber weight, when the sliding distance increases from 100 mm to 1500 mm, the wear rate increases from 2.5×10^{-6} mm to 4.5×10^{-6} mm. It suggests that longer sliding distances result in higher wear rates.

The residual plots in supplementary Fig. S1 illustrate the relationship between wear and independent variables such as filler wt%, applied load, sliding distance, and sliding velocity. For filler wt%, the residuals are randomly distributed, as shown in Fig. S1(a), with a range of -2.5 – 2.5 , which justifies the model's good fit to the data without any noticeable deviations. Similarly, Fig. S1(b) demonstrates the residuals with a range between -1.5 and 1.5 for applied load, exhibiting a slight spread but maintaining a scattered pattern, indicating the model captures the relation between applied load and wear rate most effectively. Moreover, in Figs. S1(c) and S1(d), the residuals for both sliding velocity and sliding distance fall between -2.0 and 2.5 and -3.0 and 3.0 , respectively, with no discernible trend, confirming that the model adequately represents the impact of sliding velocity and sliding distance on wear. The narrow range of residuals indicates that the model is well-

suited for explaining wear rate.

3.4.2. Coefficient of friction

The ANOVA analysis in was conducted based on a central composite design (CCD) in a quadratic model, using response surface methodology (RSM), to examine the most influential parameter out of fiber wt% (B_1), applied load (B_2), sliding velocity (B_3), and sliding distance (B_4) on the coefficient of friction presented in Table S2. The model with 7.42 and 0.0024 as F and P values indicates that the model is statistically significant, which justifies that the variation of parameters affects a substantial portion of the variance in the coefficient of friction. The applied load (B_2) and sliding distance (B_4) had the most significant effects, with p -values of 0.0371 and 0.0029, respectively. Meanwhile, with p -values 0.5355 and 0.0695, the fiber wt% (B_1) and sliding velocity (B_3) are insignificant, respectively. Among them, sliding distance (B_4) suggests a strong influence on the coefficient of friction, followed by applied load (B_2). Along with Several interactions demonstrated effects, including fiber wt% and applied load ($B_1 \times B_2$), fiber wt% square and applied load ($B_1^2 \times B_2$), with a p -value of 0.0084, and 0.0397, respectively, represent the combined effect on coefficient of friction. In contrast, interactions such as fiber wt% and sliding velocity ($B_1 \times B_3$) and sliding velocity and sliding distance ($B_3 \times B_4$) were not insignificant, with p -values of 0.2364 and 0.1226, respectively. The model has an R^2 of 92.03 %, demonstrating a strong fit, with an adjusted R^2 of 0.7963, confirming the robustness of the model even after adjusting for the number of predictors. Eq. (19) represents a multiple regression model for predicting the COF based on the interaction effects and quadratic terms of the

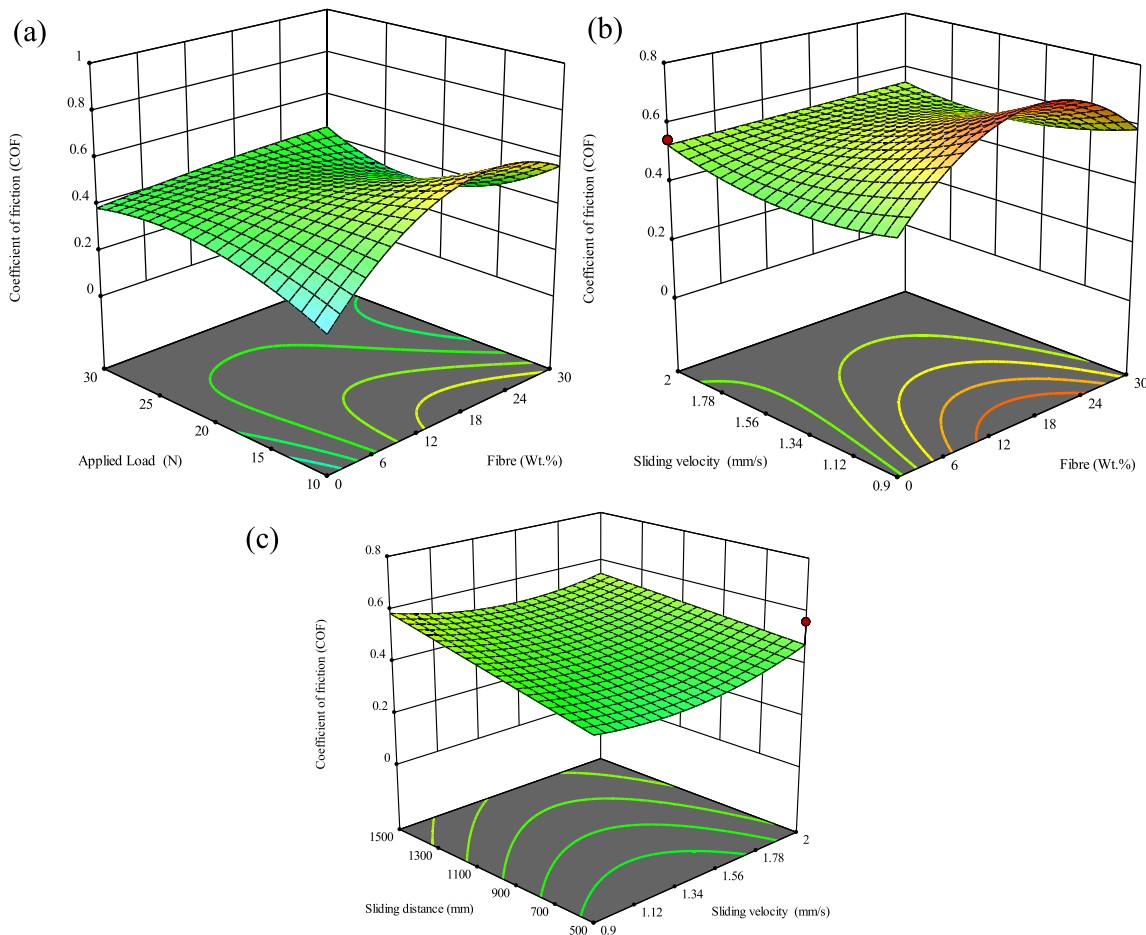


Fig. 7. 3D Surface Plots presents the effect of the coefficient of friction with (a) fiber wt% vs. applied load, (b) sliding velocity vs. fiber wt%, and (c) sliding distance vs. sliding velocity.

independent variables (B_1 , B_2 , B_3 , and B_4),

COF decreases from 0.7 to 0.4, showing that longer sliding distances reduce friction.

$$\begin{aligned} \text{Coefficient of friction (COF)} = & 0.4874 - 0.0325B_1 - 0.0978B_2 - 0.0822B_3 + 0.1627B_4 - 0.0672B_1 \times B_2 - 0.0215B_1 \times B_3 - 0.0288B_3 \\ & \times B_4 - 0.1168B_1^2 + 0.0418B_2^2 + 0.0541B_3^2 + 0.1046B_1^2 \times B_2 + 0.1117B_1^2 \times B_3 - 0.0989B_1^2 \times B_4 + 0.1302B_1 \times B_2^2 \quad \dots \quad (19) \end{aligned}$$

The 3D surface plots shown in Fig. 7 illustrate the effect of parameters such as fiber wt%, applied load, sliding velocity, and sliding distance on the coefficient of friction (COF). In Fig. 7(a), it is presented that while raising the fiber wt% from 0 % to 30 %, the COF reduced from 0.8 to 0.4, especially with higher applied load values up to 30 N, which justifies that both fiber wt% and applied load negatively affect the COF. Similarly, Fig. 7(b) shows the relationship between fiber wt% and sliding velocity (mm/s), as fiber wt% rises from 0 % to 30 %, the COF drops from 0.6 to 0.2. Still, this decrease slows down at higher sliding velocity values above 1.2 m/s, indicating that the influence of sliding velocity on COF diminishes at higher fiber wt%. Moreover, Fig. 7(c) illustrates the effect of fiber wt% and sliding distance (mm), at 30 % fiber wt%, as the sliding distance increases from 100 mm to 1500 mm, the

The residual plots in supplementary Fig. S2 illustrate the relationship between the coefficient of friction and the independent variables, including filler wt%, applied load, sliding distance, and sliding velocity. Fig. S2(a) presents the residual plots for fiber wt%, which range from approximately -2.5 – 2.5 for standardised residuals and exhibit a random distribution, indicating that the model fits the data. Similarly, the residuals for applied load, sliding velocity and sliding distance in Figs. S2(b), S2(c) and S2(d) ranges from -2 – 2 , -1.5 – 1.5 and -2.5 – 3 with a scattered distribution, suggesting that the model effectively captures the relationship between parameters (applied load, sliding velocity and sliding distance) This larger spread indicates that the model may not fully capture the relationship between parameters and the coefficient of friction, suggesting the need for further refinement.

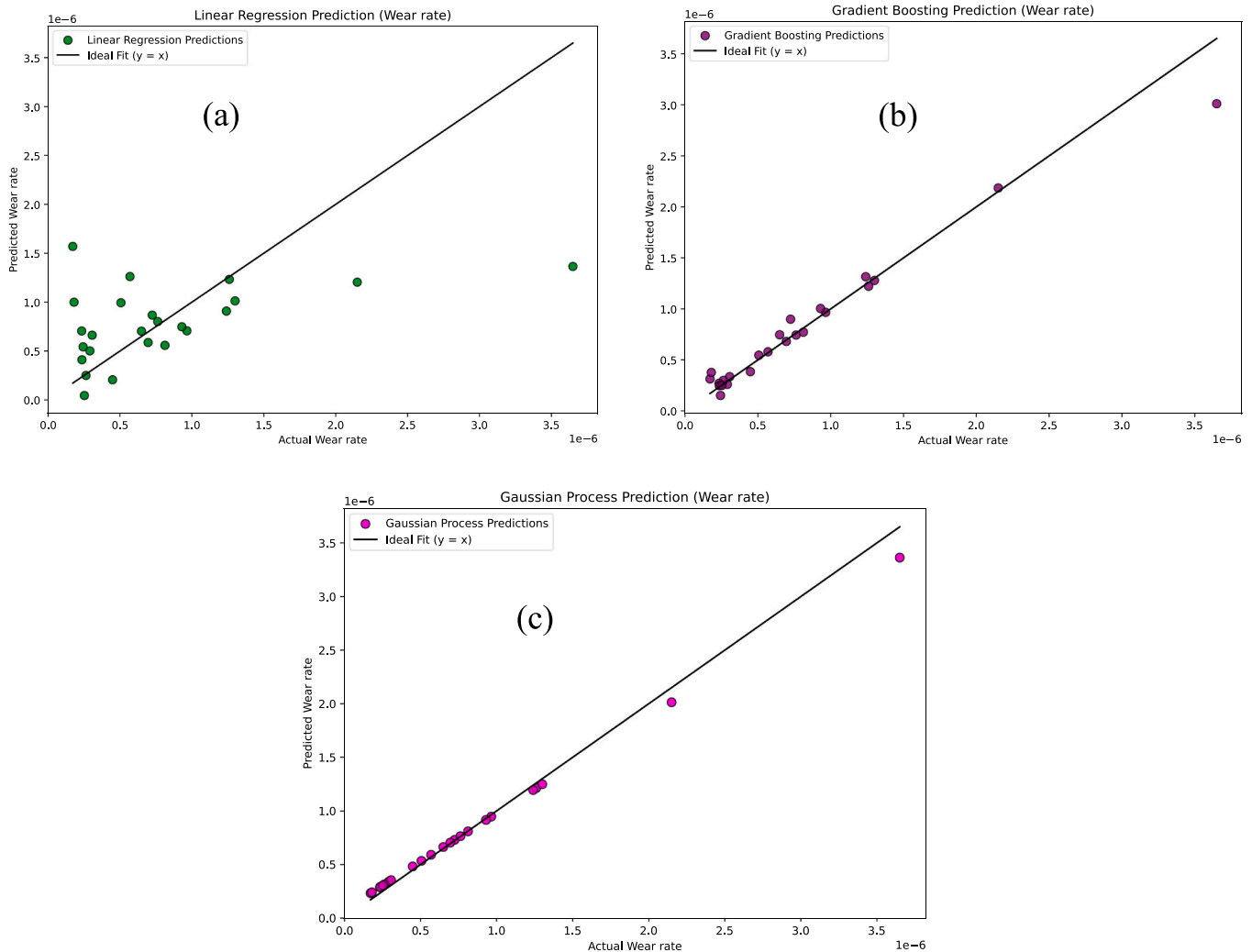


Fig. 8. Parity plots for actual vs. prediction of wear rate with (a) linear regression, (b) gradient boosting, and (c) Gaussian process.

3.5. Machine learning

3.5.1. Wear rate

The parity plots of actual versus predicted wear rate are presented in Fig. 8(a-c), with ML models, including linear regression, gradient boosting, and Gaussian process. In Fig. 8(a), the linear regression model is represented by scattered green dots that are further from the ideal fit line (black line), indicating poor prediction accuracy. This is calibrated by the error metrics: R^2 of 0.24, MSE of 4.36×10^{-13} , RMSE of 6.61×10^{-7} , MAE of 4.29×10^{-7} , MAPE of 104.41 %, and RRMSE of 84.18 %, highlighting a weak correlation with the actual values and significant errors in prediction. In Fig. 8(b), the Gradient Boosting model with the purple dots is closer to the ideal fit line, suggesting a much better performance than Linear Regression. The model achieves an R^2 of 0.96, with a significantly lower MSE of 2.26×10^{-14} , RMSE of 1.50×10^{-7} , MAE of 7.99×10^{-8} , MAPE of 16.50 %, and RRMSE of 19.17 % indicating a more consistent prediction across the data. In Fig. 8(c), the Gaussian Process model has a superior predictive accuracy, with the magenta dots aligned perfectly closer to the ideal line, with error matrices of R^2 of 0.99, the lowest MSE of 5.74×10^{-15} , RMSE of 7.57×10^{-8} , MAE of 5.02×10^{-8} , and MAPE of 10.97 %, suggesting that it

provides highly accurate predictions with minimal error. Its RRMSE of 9.65 % conforms to accuracy and consistency. The Gaussian Process outperforms with the highest R^2 and low error matrices compared with both Gradient Boosting and Linear Regression in terms of predictive accuracy for wear rate (Liu et al., 2025).

The confusion matrix of wear rate obtained from machine learning models such as linear regression, gradient boosting, and Gaussian process is shown in Fig. 9(a-c). It divides the wear rate into four categories, including Very Low, Low, Medium, and High, to evaluate how well different methods can predict wear rate at varying levels. The linear regression model in Fig. 9(a) exhibits one high-to-high mapping, one moderate-to-moderate, 4 low-to-low, and 4 very low-to-very low. A total of 10 out of 24 were predicted with an accuracy of 41.67 %. This lower accuracy can be attributed to its simplicity, as it is unable to capture the complex relationships between the data features and the wear rate. In contrast, Fig. 9(b) presents the Gradient Boosting method, which achieved three high and five moderates, low, and very low, for a total of 19 out of 24, with an accuracy of 79.17 %, demonstrating better performance in predicting wear rate categories. Similarly, Fig. 9(c) illustrates that the Gaussian Process method performed the best, with two high, 6 moderate, 6 low, and 6 very low, for a total of 20 out of 24 samples, with

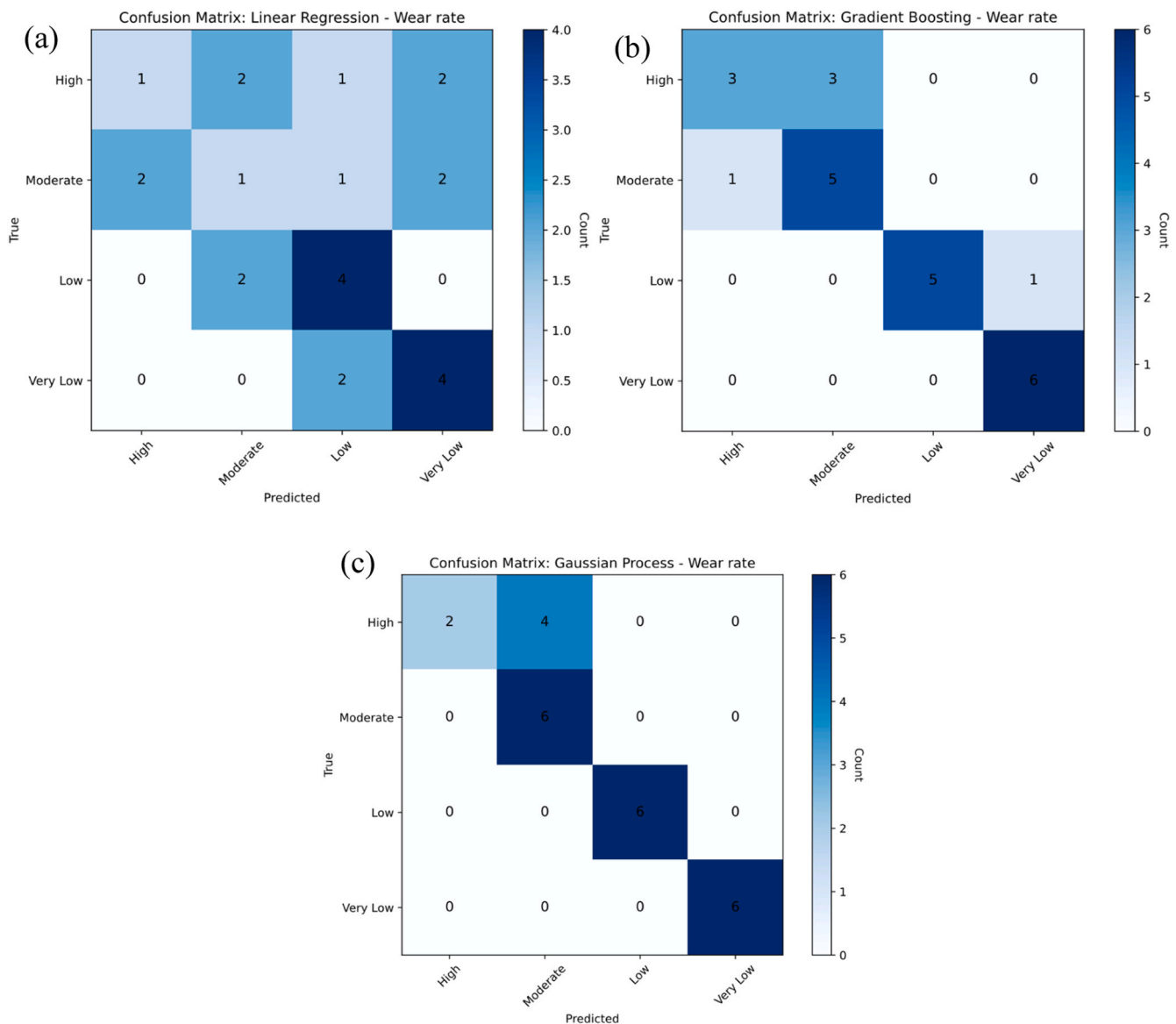


Fig. 9. Confusion matrix for wear rate with (a) linear regression, (b) gradient boosting, and (c) Gaussian process.

an accuracy of 83.33 %, indicating its strong ability to predict the wear levels accurately. This categorization helped us assess the effectiveness of each method in real-world scenarios, where precise wear rate prediction is crucial for maintenance decision-making (Danish et al., 2024).

Supplementary Fig. S3 illustrates the SHAP analysis for the prediction of wear rate by various machine learning models, such as linear regression, gradient boosting, and Gaussian process, to observe the influence of features, including fiber wt%, applied load, sliding distance, and sliding velocity. The ShAP value plots in Figs. S3(a), S3(c), and S3(e) depict the influence of each feature on the wear rate, with the color intensity represented by the SHAP values. Similarly, the bar charts in Figs. S3(b), S3(d), and S3(f) display the mean absolute SHAP values, which offer a comparative view of feature importance. The linear regression model in Fig. S3(a) demonstrated that the sliding distance is the most influential feature, with ShAP values ranging from -0.4 – 0.4 , demonstrating its substantial impact on the prediction, which is justified by the feature importance plot in Fig. S3(b), confirming the highest importance score of around 0.25. Applied load and fiber content also contribute to the prediction, though with more minor impacts, while sliding velocity has the least influence. Similarly, for the Gradient Boosting model in Fig. S3(c), the resulting higher ShAP values range from -1 – 1 for fiber wt%. This indicates an increasing fiber wt% significantly affects the wear rate. The feature importance plot in Fig. S3 (d) justifies that fiber wt% is the most important feature with a score of 0.4, followed by sliding distance at 0.3, while applied load also plays a

role. Finally, for the Gaussian Process model shown in Fig. S3(e), fiber wt% is more dominant, with ShAP values spanning from -0.75 to $+0.75$, reflecting its substantial effect on wear-rate prediction. Sliding distance remains significant but slightly less influential, and applied load sees a slight decrease in importance. The feature importance plot in Fig. S3(f) highlights fiber content as the most crucial feature (0.4), with sliding distance at 0.3 and applied load at 0.25. The sliding distance consistently plays a central role across all models, but the increasing complexity of the models reveals deeper interactions, emphasizing the growing significance of fiber wt% and applied load for wear behavior.

3.5.2. Coefficient of friction

The parity plots in Fig. 10 (a-c) compare the actual and predicted coefficient of friction (COF) obtained from machine learning models, including linear regression, gradient boosting, and Gaussian process. The linear regression model in Fig. 10(a), represented by green dots, shows an apparent deviation from the ideal fit line (the black line), indicating poor predictive accuracy, with an R^2 of 0.24, an MSE of 0.0228, an RMSE of 0.1509, and an MAE of 0.1263. MAPE is 45.84 %, and the RRMSE is 30.93 %. Similarly, Fig. 10(b) shows the Gradient Boosting model, represented by purple dots, which aligns much more closely with the ideal fit line, suggesting improved prediction accuracy over linear regression, with R^2 of 0.95, MSE of 0.0014, and RMSE of 0.0375. The MAE of 0.0288, MAPE of 7.46 %, and RRMSE of 7.69 % demonstrate the model's enhanced ability to capture the underlying

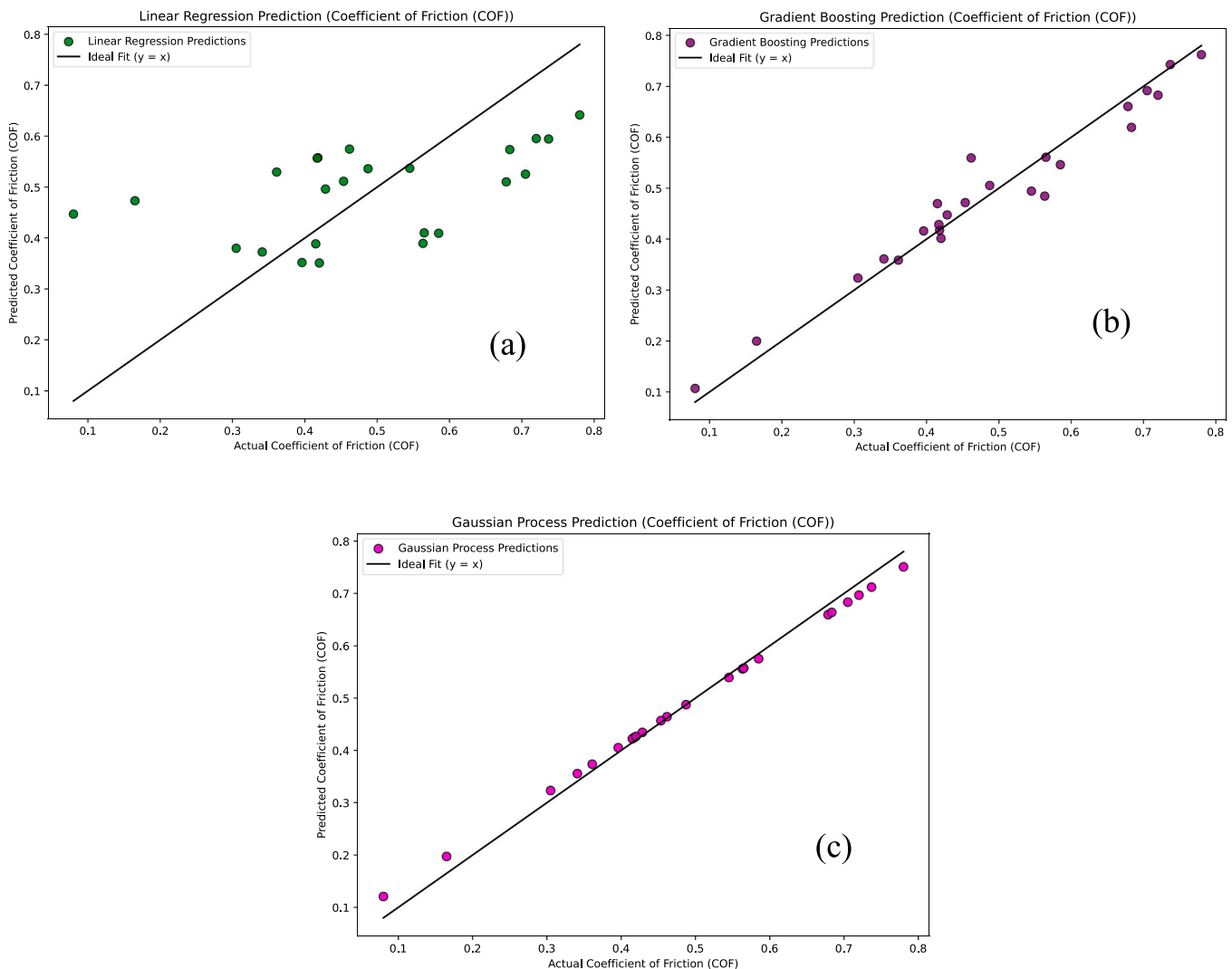


Fig. 10. Parity plots for actual vs. prediction of the coefficient of friction with (a) linear regression, (b) gradient boosting, and (c) Gaussian process.

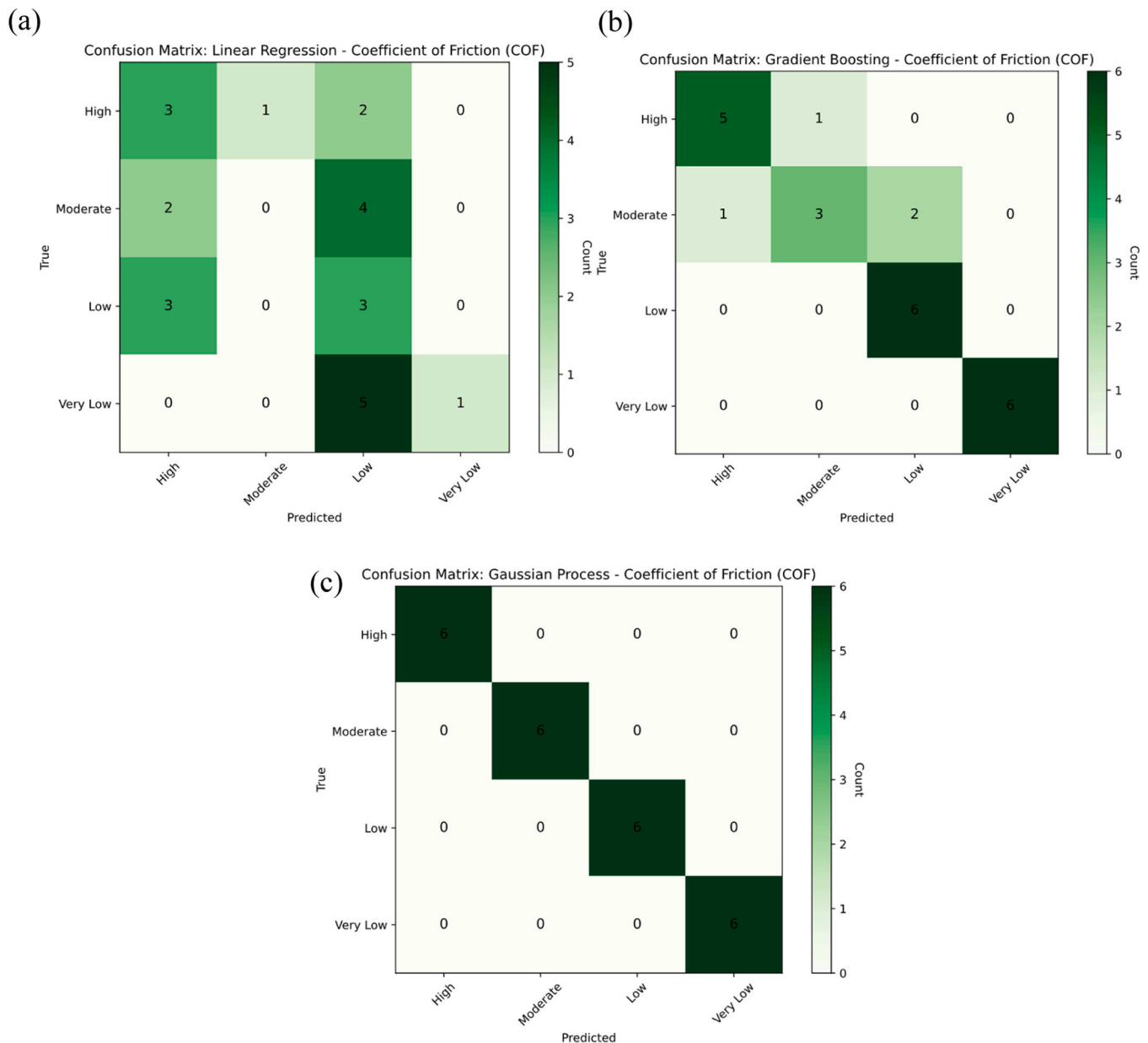


Fig. 11. Confusion matrix for coefficient of friction with (a) linear regression, (b) gradient boosting, and (c) Gaussian process.

patterns, offering a better balance between prediction accuracy and computational efficiency. Finally, in Fig. 10(c), the Gaussian Process model, depicted by magenta dots, aligns almost perfectly with the ideal fit line, reflecting superior predictive accuracy. With R^2 of 0.99, the lowest MSE of 0.0003, RMSE of 0.0174, and MAE of 0.0140, the model delivers highly accurate predictions. The MAPE of 5.03 % and RRMSE of 3.56 % further confirm the model's minimal error and excellent consistency in its predictions. The Gaussian Process model outperforms both Gradient Boosting and Linear Regression in terms of predictive accuracy, achieving the highest R^2 and the lowest error metrics across the board. However, gradient boosting also offers strong performance, balancing computational efficiency with high accuracy, while linear regression, with its higher errors, proves less suitable for accurate COF prediction (Liu et al., 2025).

Fig. 11 (a-c) shows the confusion matrix results that compare the experimentally obtained COF with the COF obtained from various machine learning models such as linear regression, gradient boosting, and Gaussian process is classified into very low, low, moderate, and high to

correlate to analyze the prediction accuracy of the model. The linear regression model in Fig. 11(a) achieved an actual position of three as high, three as low, and one as very low, for a total of 7 out of 24, which were predicted with an accuracy of 29.17 %. In contrast, the gradient boosting model in Fig. 11(b) achieved an actual position of five high, three moderate, six low, and six very low for a total of 20 out of 24 samples with a higher accuracy of 79.17 %, demonstrating moderate performance in predicting the COF categories. Similarly, Fig. 11(c) illustrates that the Gaussian Process method outperformed both, with a perfect accuracy of 100 %, indicating its strong ability to predict the COF levels accurately. This categorization helped us assess the effectiveness of each method in real-world scenarios, where precise friction prediction is crucial for material selection and design (Danish et al., 2024).

The SHAP (Shapley Additive exPlanations) analysis is presented in supplementary Fig. S4, illustrating the influence of factors on COF prediction for the machine learning models, including linear regression, gradient boosting, and Gaussian Process. The linear regression model in

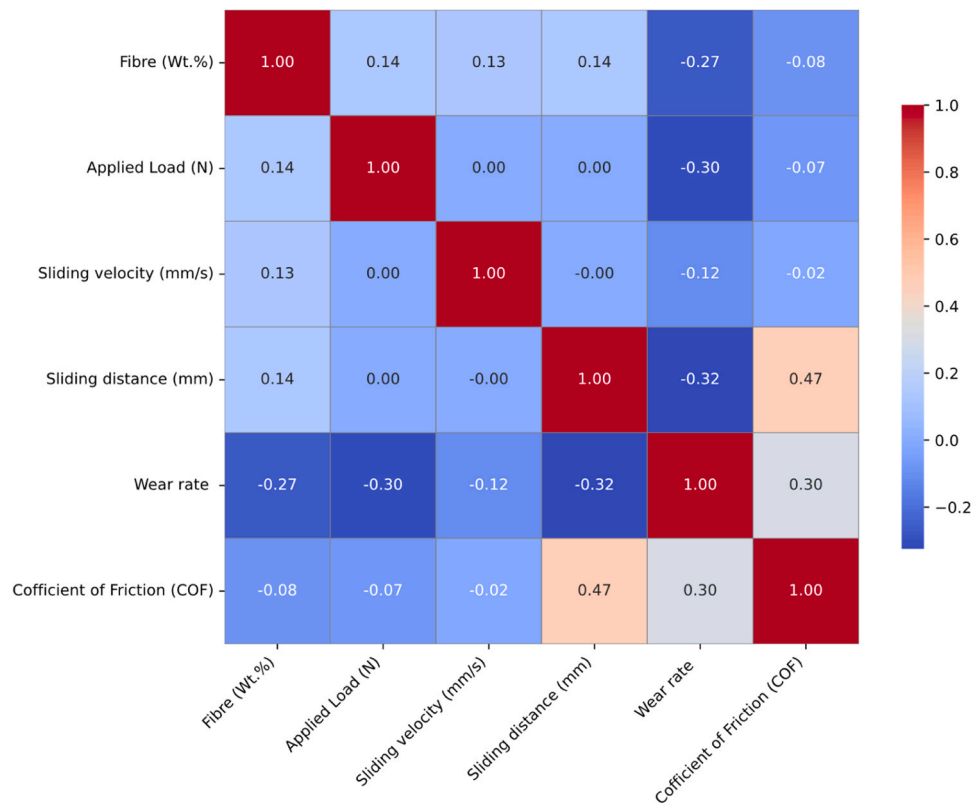


Fig. 12. Correlation heat map between actual and predicted values.

Fig. S4(a) shows that the sliding distance has a stronger influence with higher ShAP values between -0.8 and $+0.6$, which is the most important feature, affecting the COF. Meanwhile, the feature importance plot in Fig. S4(b) also shows that sliding distance is the most important feature, with a score of about 0.4. The wt% of fiber is also significant, but the applied load and the sliding velocity have less effect. The gradient boosting model in Fig. S4(c) and Fig. S4(d) presents the SHAP values ranging from -1.5 – 1.5 , which have the most impact on the model's predictions, especially for fiber content, and the feature importance plot represents that fiber wt% is the most important feature, with a score of 0.6, followed by sliding distance, with a score of 0.3. Applied load and sliding velocity are less critical. Similarly, in Fig. S4(e) and Fig. S4(f) for the Gaussian process model, the fiber wt% has a significant influence on the coefficient of friction, with SHAP values ranging from -1.0 to $+1.0$. The sliding distance is essential, but it has a minimal impact. The applied load and sliding velocity still have less of an effect, while the feature importance plot shows that the most important feature is fiber content (0.6), followed by sliding distance (0.3), applied load (0.2), and sliding velocity (0.1). In all models, the sliding distance is the most critical factor. The models are getting more complicated, which shows deeper interactions. This is especially true for gradient boosting and Gaussian process models, where fiber wt% is becoming more critical.

The Pearson correlation heatmap in Fig. 12 shows the relationships between fiber content, applied load, sliding velocity, sliding distance, wear rate, and coefficient of friction (COF). The color gradient, ranging from red (strong positive correlation) to blue (strong negative correlation), visualizes the strength of these relationships. Fiber content is weakly positively correlated with applied load 0.14, sliding velocity 0.13, and sliding distance 0.14, all reflected in lighter blue shades. It is moderately negatively correlated with wear rate (-0.27), represented by a medium blue color, and very weakly negatively correlated with COF (-0.08), indicated by a very light blue. Applied load shows a positive correlation with fiber content (0.14) but a moderate negative correlation with wear rate (-0.30), as shown in light blue. Sliding velocity has

minimal influence on wear rate -0.12 and COF -0.02 , with weak blue hues. Sliding distance is positively correlated with both wear rate 0.47 and COF 0.47, represented by a deeper shade of red, indicating moderate positive relationships. Wear rate is negatively correlated with fiber content -0.27 and applied load -0.30 , depicted in medium blue. In contrast, it is positively correlated with sliding distance 0.47 and COF 0.30, shown in red and light red. COF exhibits a moderate positive correlation with sliding distance 0.47 and wear rate 0.30, marked in deeper red shades. The heatmap highlights weak to moderate correlations among the variables, with the most significant relationships observed between sliding distance and COF, as well as between wear rate and sliding distance, indicated by the darker red and blue color intensities.

Supplementary Fig. S5(a) compares the experimentally measured wear rate with the values predicted using ANOVA and various machine learning models. Across the 24 test conditions (S1–S24), the wear rate varies substantially, ranging from 0.18 to 3.65 mm³/N·mm. The deviation analysis highlights the Gaussian process model as the most accurate predictor—for instance, in test condition S1, it shows only a 2.59 % error. Gradient Boosting follows with moderate deviations (5.46 %), while Linear Regression exhibits significantly larger errors. Although ANOVA is less accurate than the ML models, it still provides reasonably close estimates in cases such as test conditions S2 and S3. Supplementary Fig. S5(b) presents the comparison for the coefficient of friction (COF), which varies from 0.18 to 0.75. The Gaussian Process model again demonstrates the best performance, with a deviation of just 0.35 % in S1. Gradient boosting performs moderately, whereas linear regression shows higher discrepancies. ANOVA produces moderate errors (e.g., 17.54 % in test condition S19) but effectively captures overall trends. The Gaussian Process is the most reliable model, followed by Gradient Boosting, while ANOVA remains a strong, simpler alternative for trend-based prediction.

4. Conclusion

This study demonstrates that reinforcement of epoxy composites with pineapple leaf fiber significantly improves their tribological performance. The fiber weight fraction was found to be the most influential factor in enhancing wear resistance and reducing the coefficient of friction. The optimized composite exhibited superior wear resistance and frictional properties, confirming the potential of pineapple leaf fiber as a sustainable reinforcement for epoxy-based materials. Machine learning models, particularly Gaussian Process regression, provided accurate predictions for tribological performance, supporting the optimization process with minimal experimental effort. These findings highlight the feasibility of using bio-based fibers to enhance the mechanical properties of composites for industrial applications.

CRediT authorship contribution statement

Joy Djuansjah: Writing – review & editing, Visualization, Investigation, Formal analysis. **Eng Eng Ngu:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Santosh Kumar Sahu:** Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Sundarasetty Harishbabu:** Visualization, Software, Investigation, Formal analysis, Conceptualization. **It Ee Lee:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Nadir Ayrimis:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Borhen Louhichi:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.indcrop.2026.122865](https://doi.org/10.1016/j.indcrop.2026.122865).

Data availability

Data will be made available on request.

References

- Chavhan, G., Wankhade, L., 2020. Optimization of test parameters that influence on dry sliding wear performance of steel embedded glass/epoxy hybrid composites by using the taguchi approach. *Tribol. Ind.* 42, 556. <https://doi.org/10.24874/ti.863.03.20.09>.
- Chee, S.S., Jawaid, M., Sultan, M.T.H., Alotman, O.Y., Abdullah, L.C., 2019. Accelerated weathering and soil burial effects on colour, biodegradability and thermal properties of bamboo/kenaf/epoxy hybrid composites. *Polym. Test.* 79, 106054. <https://doi.org/10.1016/j.polymertesting.2019.106054>.
- Danish, M., Gupta, M.K., Irfan, S.A., Ghazali, S.M., Rathore, M.F., Krolczyk, G.M., Alsaady, A., 2024. Machine learning models for prediction and classification of tool wear in sustainable milling of additively manufactured 316 stainless steel. *Results Eng.* 22, 102015. <https://doi.org/10.1016/j.rineng.2024.102015>.
- Dharmalingam, S., Sasikumar, T., 2025. Mechanical, wear, and flammability properties of silanized cow dung biosilica-dispersed corn husk Fiber-reinforced epoxy composites. *Biomass. Conv. Bioref.* 15, 1169–1179. <https://doi.org/10.1007/s13399-023-05082-z>.

- Feng, Y., et al., 2026. A novel strategy for oil storage and transport in epoxy composites using biomass *Juncus effusus* fibers. *Surf. Interfaces* 80, 108188. <https://doi.org/10.1016/j.surf.2025.108188>.
- Feng, Y., Pan, B., Liu, H., Zhou, Y., Ding, X., Yuan, X., 2024. Mechanical and tribological behavior of graphene oxide supported aramid fiber reinforced epoxy resin. *J. Polym. Mater.* 41, 3. <https://doi.org/10.32604/jpm.2024.055558>.
- Giridharan, K., Elumalai, B., Vinothkumar, M., et al., 2025. Investigation of dry sliding wear and mechanical properties of hybrid epoxy composites reinforced with pineapple leaf and roselle fibers. *Sci. Rep.* 15, 27991. <https://doi.org/10.1038/s41598-025-10431-1>.
- Harishbabu, S., Alrasheedi, N.H., Louhichi, B., Sahu, S.K., Ma, Q., 2025a. Optimization and prediction of mechanical properties of additively manufactured PLA/GNP composites via response surface methodology and machine learning models. *Polymers* 17, 2894. <https://doi.org/10.3390/polym17212894>.
- Harishbabu, S., Alrasheedi, N.H., Louhichi, B., Sreekanth, P.S.R., Sahu, S.K., 2025b. Machine learning-assisted synergistic optimization of 3d printing parameters for enhanced mechanical properties of pla/boron nitride nanocomposites. *Machines* 13, 949. <https://doi.org/10.3390/machines13100949>.
- Ibrahim, M.M., El-Tayeb, N., Shazly, M., El-Sayed Seleman, M., 2024. An experimental study on the effect of graphite microparticles on the mechanical and tribological properties of epoxy matrix composites. *Funct. Compos. Mater.* 5, 2. <https://doi.org/10.1186/s42252-023-00046-y>.
- Jagadeesan, N., Selvaraj, A., Nagaraja, S., et al., 2022. Response Surface Methodology Based Optimization of Test Parameter in Glass Fiber Reinforced Polyamide 66 for Dry Sliding, Tribological Performance. *Materials* 15, 6520. <https://doi.org/10.3390/ma15196520>.
- Kumar, R., Anand, A., 2019. Tribological behavior of natural fiber reinforced epoxy based composites: A review. *Mater. Today Proc.* 18, 3247–3251. <https://doi.org/10.1016/j.matpr.2019.07.200>.
- Kumar, T.S., Raghu, R., Priyadharshini, G.S., et al., 2024. A study on microstructural, mechanical properties, and optimization of wear behaviour of friction stir processed AZ31/TiC composites using response surface methodology. *Sci. Rep.* 14, 18729. <https://doi.org/10.1038/s41598-024-69348-w>.
- Lakshmi, A.R., Satishkumar, P., Nagabhooshanam, N., et al., 2025. Shear strength, wear, thermal conductivity, and hydrophobicity behavior of fox millet husk biosilica and Amaranthus dubius stem fiber-reinforced epoxy composite: a concept of biomass conversion. *Biomass. Conv. Bioref.* 15, 2893–2901. <https://doi.org/10.1007/s13399-023-04854-x>.
- Liu, S., Du, F., Gao, Z., Han, X., Shen, Y., Wang, J., Wang, W., 2025. Research on Sliding Bearing Wear Prediction Based on Gaussian Process Regression. *J. Mater. Eng. Perform.* 1–12. <https://doi.org/10.1007/s11665-025-12924-6>.
- Mahmud, Md, Rabbi, Z.A., Islam, S.M.F., Hossain, N, Md.D., 2025. Synthesis and applications of natural fiber-reinforced epoxy composites: A comprehensive review. *SPE Polym.* 6, e10161. (Article).
- Maleki, B., Ghamari Kargar, P., Sedigh Ashrafi, S., 2025. Biobased composites: A sustainable pathway for modern applications. *Mater. Sci.* 8 IntechOpen. <https://doi.org/10.5772/intechopen.1009464>.
- Manoj, E., Selvakumar, G., Sivaprakash, V., et al., 2025. Fabrication and Characterization of Porous Biochar and Si₃N₄ Bioceramic Toughened Natural Fiber Epoxy Composite. *Silicon* 17, 2417–2428. <https://doi.org/10.1007/s12633-025-0335-2>.
- Muralidhara, B., Babu, S.P.K., Suresha, B., Yogesha, K.K., Vasundhara, M.G., Kalavathi, G.K., Goudar, D.M., Kurahatti, R.V., Bhat, S.K., 2025. Enhanced mechanical, thermal, and wear performance of halloysite nanotube infused carbon fiber epoxy composites. *Sci. Rep.* 15, 30019. <https://doi.org/10.1038/s41598-025-15142-1>.
- Nagaraja, S., Anand, P.B., H.D., S., Ammarullah, M.I., 2024. Influence of fly ash filler on the mechanical properties and water absorption behaviour of epoxy polymer composites reinforced with pineapple leaf fiber for biomedical applications. *RSC Adv.* 14, 14680–14696. <https://doi.org/10.1039/d4ra00529e>.
- Pernica, J., Vodák, M., Šarocký, R., et al., 2022. Mechanical Properties of Recycled Polymer Materials in Additive Manufacturing. *Manuf. Technol.* 22, 200–203. <https://doi.org/10.21062/mft.2022.017>.
- Qiuqian, W., GaoMin, Z.K., Chenchen, 2025. A light gradient boosting machine learning-based approach for predicting clinical data breast cancer. *Multiscale Multidiscip. Model. Exp. Des.* 8, 75. <https://doi.org/10.1007/s41939-024-00662-6>.
- Salguero, J., Iglesias, P., Vazquez-Martinez, J.M., Batista, M., Del Sol, I., 2022. A comparative study of disk wear volume evaluation of al2024 based on ASTM G99. October 30–November 3, 2022. V009T15A002. In: Proceedings of the ASME 2022 International Mechanical Engineering Congress and Exposition, 9. ASME, Columbus, Ohio, USA. <https://doi.org/10.1115/IMECE2022-94625>.
- Saran, C.S., Satapathy, A., 2025. Parametric analysis and prediction of dry adhesive wear characteristics of waste glass dust filled hemp-epoxy composites. *J. Adhes. Sci. Technol.* 39, 1181–1199. <https://doi.org/10.1080/01694243.2024.2441890>.
- Sidh, K.N., Hirani, H., 2025. Enhanced wear volume analysis using three-dimensional profilometry for four-ball tribometer setups. *J. Tribol.* 147, 091110. <https://doi.org/10.1115/1.4068170>.
- Silhavy, R., Silhavy, P., Prokopova, Z., 2017. Analysis and selection of a regression model for the use case points method using a stepwise approach. *J. Syst. Softw.* 125, 1–14. <https://doi.org/10.1016/j.jss.2016.11.029>.
- Singh, A., Chaitanya, S., Mohal, S., et al., 2025. Wear behavior optimization of friction stir processed Mg-TiC composites through response surface methodology. *Appl. Phys.* A 131, 798. <https://doi.org/10.1007/s00339-025-08838-8>.
- Singh, T., Gangil, B., Ranakoti, L., Joshi, A., 2021. Effect of silica nanoparticles on physical, mechanical, and wear properties of natural fiber reinforced polymer composites. *Polym. Compos* 42, 2396–2407. <https://doi.org/10.1002/pc.25986>.

- Skosana, S.J., Khoathane, C., Malwela, T., 2025. Driving towards sustainability: A review of natural fiber reinforced polymer composites for eco-friendly automotive lightweighting. *J. Thermoplast. Compos. Mater.* 38, 754–780. <https://doi.org/10.1177/08927057241254324>.
- Sumesh, K.R., Ajithram, A., Anjumol, K.S., Sai Krishnan, G., 2024. Influence of natural fiber addition and fiber length in determining the wear resistance of epoxy-based composites. *Polym. Compos* 45, 3029–3042. <https://doi.org/10.1002/pc.27968>.
- Sundarasetty, H., Louhichi, B., Alrasheedi, N.H., et al., 2025. Machine learning guided process optimization and sustainable valorization of coconut biochar filled PLA biocomposites. *Sci. Rep.* 15, 34647. <https://doi.org/10.1038/s41598-025-19791-0>.
- Sundarasetty, H., Sahu, S.K., 2025. Tribological behavior of PLA reinforced with boron nitride nanoparticles using Taguchi and machine learning approaches. *Results Eng.* 26, 104772. <https://doi.org/10.1016/j.rineng.2025.104772>.
- Suresha, B., Chandramohan, G., Siddaramaiah, Samapthkumaran, P., Seetharamu, S., 2007. Three-body abrasive wear behaviour of carbon and glass fiber reinforced epoxy composites. *Mater. Sci. Eng. A* 443, 285–291. <https://doi.org/10.1016/j.msea.2006.09.016>.
- Tasgin, Y., Demircan, G., Kandemir, S., Acikgoz, A., 2024. Mechanical, wear and thermal properties of natural fiber-reinforced epoxy composite: Cotton, sisal, coir and wool fibers. *J. Mater. Sci.* 59, 10844–10857. <https://doi.org/10.1007/s10853-024-09810-2>.
- Ul-Hamid, A., 2021. Synthesis, microstructural characterization and nanoindentation of Zr, Zr-nitride and Zr-carbonitride coatings deposited using magnetron sputtering. *J. Adv. Res.* 29, 107–119. <https://doi.org/10.1016/j.jare.2020.11.010>.
- Venkatesh, R., Ballal, S., Krishnan, A.M., et al., 2023. Effect of fiber layer formation on mechanical and wear properties of natural fiber filled epoxy hybrid composites (Article). *Heliyon* 9, e16183. <https://doi.org/10.1016/j.heliyon.2023.e15934>.
- Zhang, X., Sun, X., Lin, Z., 2025. Improving soil moisture prediction using Gaussian process regression. *Smart Agric. Technol.* 11, 100905. <https://doi.org/10.1016/j.atech.2025.100905>.